

Applications of Statistical Modeling:

Module #2, Lecture #1

Structural Equation Modeling: An Overview

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What is a Structural Equation Model (SEM)?

- A statistical model used in many disciplines (especially the social sciences) to test hypotheses about causal relationships between many variables
- The models can include **observed variables** in a survey data set, in addition to **latent variables** that were not directly observed but rather *indicated* by other observed variables

What is a Structural Equation Model (SEM)?

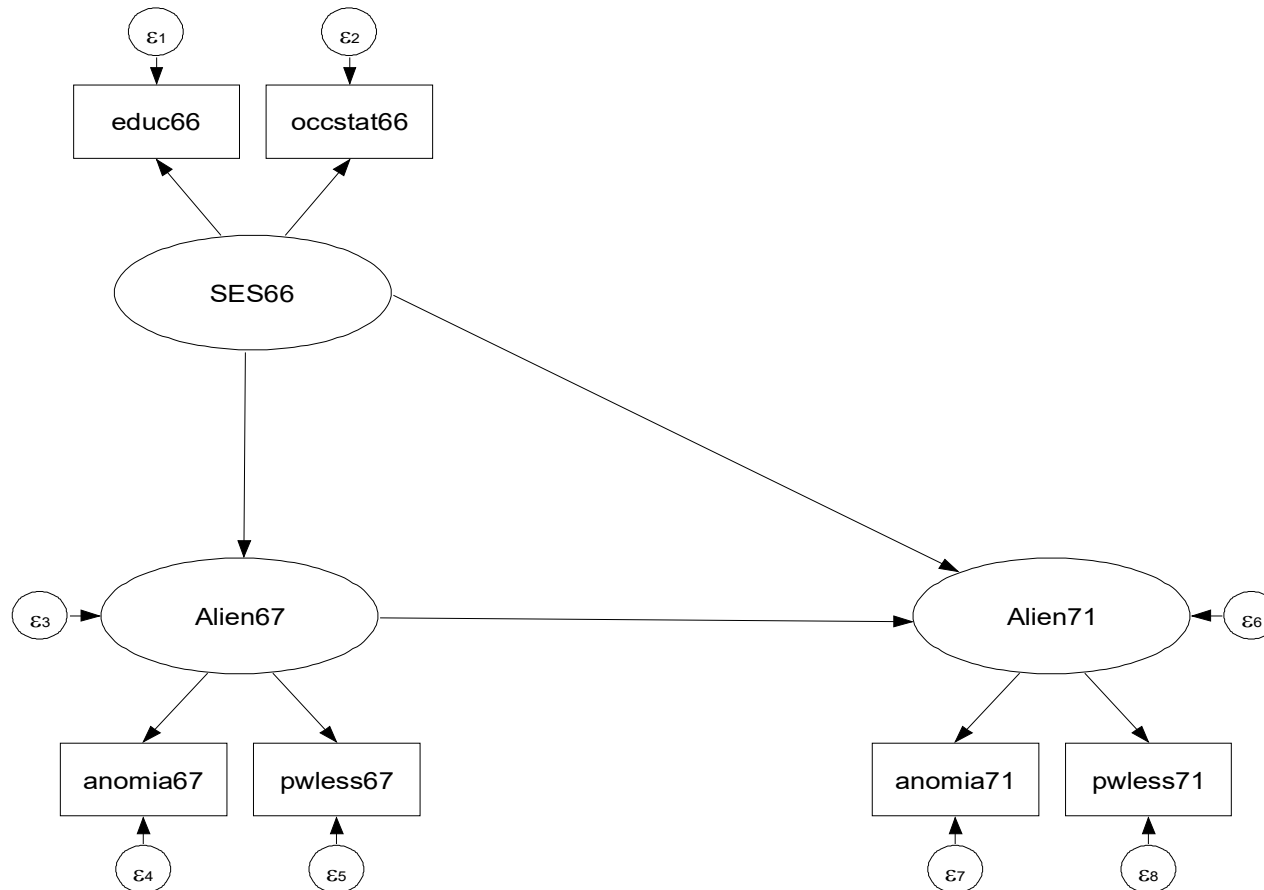
- Full SEMs generally include two components:
 - A **measurement model**, where latent variables are indicated by multiple observed variables, and
 - A **structural model**, which describes the relationships between both latent and observed variables of interest
- There are many types of models that fit into the general SEM framework!

What is a Structural Equation Model (SEM)?

- Common examples of SEMs:
 - **Path Models**: specify causal relationships between observed variables (e.g., mediation models)
 - **Confirmatory Factor Analysis (CFA) Models**: The measurement portion(s) of an SEM
 - **Multilevel Models**: random coefficients are the latent variables (in a longitudinal framework)
 - **Latent Class Models**: Group cases into latent classes based on multiple indicator variables
 - **Many others!**

What do SEM equations look like?

- Consider this SEM (Wheaton et al. 1977):



The General SEM Model

- In general, an SEM model is written as follows:

Latent variable (structural) equation:

$$\eta = \alpha + B\eta + \Gamma\xi + \zeta$$

A given vector of endogenous LVs is a parametric function of other LVs, plus errors

Measurement equations:

$$x = v_x + \Lambda_x\xi + \delta$$

$$y = v_y + \Lambda_y\eta_1 + \varepsilon$$

$$z = v_z + \Lambda_z\eta_2 + \gamma$$

Vectors of observed variables are parametric functions of intercepts, the LVs, and errors

Back to the Example Model

- In our example two slides back, we have:

Structural Model:

$$Alien67 = \alpha_1 + \gamma_{11}SES66 + \varepsilon_3$$

$$Alien71 = \alpha_2 + \gamma_{21}SES66 + \gamma_{22}Alien67 + \varepsilon_6$$

Measurement Model:

$$educ66 = v_1 + \lambda_{11}SES66 + \varepsilon_1$$

$$occstat66 = v_2 + \lambda_{21}SES66 + \varepsilon_2$$

$$anomia67 = v_3 + \lambda_{31}Alien67 + \varepsilon_4$$

$$pwless67 = v_4 + \lambda_{41}Alien67 + \varepsilon_5$$

$$anomia71 = v_5 + \lambda_{51}Alien71 + \varepsilon_7$$

$$pwless71 = v_6 + \lambda_{61}Alien71 + \varepsilon_8$$

What is a Structural Equation Model (SEM)?

- SEMs use observed covariances (or correlations) in a data set to estimate the parameters of a hypothesized model
- SEMs should **not** be used for exploratory data analysis; **always start with a theory!**
- SEMs enable examinations of the **plausibility** of hypothesized models
- If a model fits well (i.e., the implied correlations match the observed correlations), it is **plausible**, but there may well be equivalent or better models

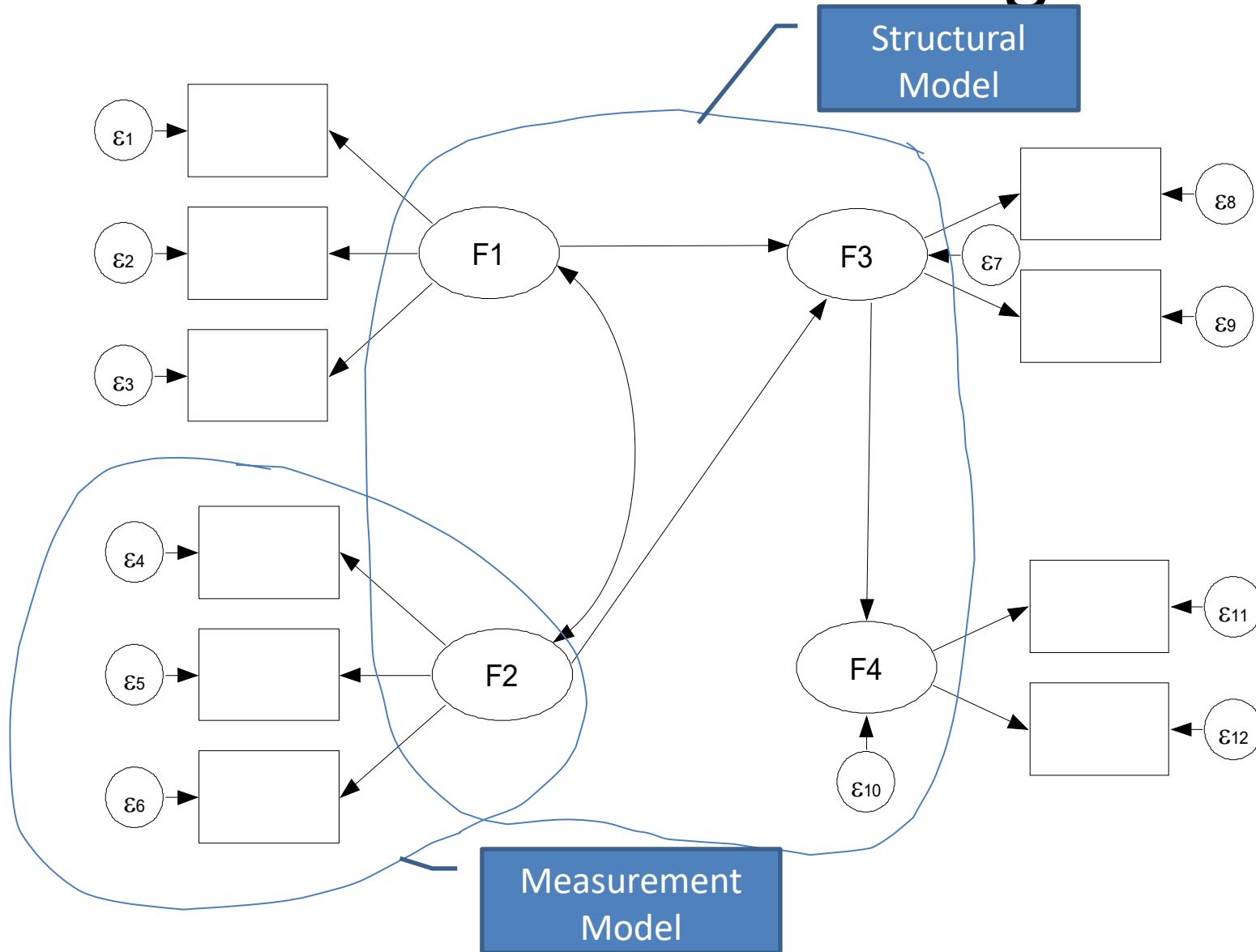
Terminology

- **Measured variables:** Those variables that are directly observed in the data set
- **Latent / factor variables:** Unobserved variables that are indicated by other variables in the data set (e.g., mathematics ability)
- **Endogenous variables:** Variables predicted by other variables (like dependent variables)
- **Exogenous variables:** Variables that only predict other variables

Conventions for Drawing SEMs

- **Squares or boxes:** observed variables available in the data set
- **Circles or ovals:** latent variables that might be indicated by other observed variables
- **Straight arrows:** indicate direct causal relationships where one variable predicts another
- **Two-sided curved arrows:** indicate covariances, generally between latent variables
- Predicted variables usually also have circles with straight arrows pointing to them, or **errors**

Conventions for Drawing SEMs



Estimation of SEMs

- **Input information:** sample covariances (or correlations) between observed variables
- Sample means are also used if there are missing data in the data set
- The observed covariance matrix is a function of a set of parameters, which define the SEM
- The covariances are **decomposed** into the parameters that describe a given model

Scaling Latent Variables

- Because latent variables are unobserved, we need to place constraints on the model prior to estimation that define the scale of all latent variables, so that models are **identified**
- One of two options are generally employed:
 - Fix the path from the latent variable to one of its indicators to be equal to 1, with 0 intercept (see **Slide 26**)
 - Fix the variance of the latent variable to be 1 (we will see an example of this at the end of the lecture)

How are Estimates Computed?

- Taken together, the parameters of an SEM combine to ***imply*** a set of means and covariances for the observed variables (**we will see several examples!**)
- Given:
 - distributional assumptions for observed variables, and
 - ***implied*** means, variances and covariances for observed variables, based on the SEM parameters,
- We can define a likelihood function, and then use maximum likelihood estimation methods

How are Estimates Computed?

- General ML fitting function / log-likelihood (MVN):

$$F_{ML} = \ln |\Sigma(\theta)| + tr(S\Sigma^{-1}(\theta)) - \ln |S| - (p + q) \\ + (\bar{z} - \mu(\theta))' \Sigma^{-1}(\theta) (\bar{z} - \mu(\theta)) \}$$
 if means estimated

where

θ = vector of all parameters to be estimated

$\Sigma(\theta)$ = ***implied*** covariance matrix for variables

S = sample covariance matrix for variables

$p + q$ = number of observed variables

$\bar{z}$ = vector of sample means for observed variables

$\mu(\theta)$ = vector of implied means based on model

How are Estimates Computed?

- We take the derivative of this function w.r.t. θ , and find estimates that set the function equal to zero (iterative numerical algorithms)
- We seek estimates of the SEM parameters that **minimize** the differences between the implied and sample covariances (and means)!
- Exact forms of the fitting function will vary depending on distributional assumptions and the estimation method used; examples soon!

Additional Tools for Inference

- Overall Chi-square test of model fit:
 - Compute a Chi-square statistic, based on the differences between the sample covariance matrix and the implied covariance matrix
 - **Degrees of Freedom** (DF) are equal to the difference in the number of unique elements in the input covariance matrix [$k(k+1) / 2$, for k observed variables] (plus the number of unique means, if applicable), and the number of parameters estimated
- Likelihood ratio tests (more shortly)

Model Fit Criteria

- **CFI:** Compare fitted model to null model assuming no relationships between variables; how much better is fitted model? Values greater than 0.9 or 0.95 desired
- **RMSEA:** Root mean squared error of approximation; a test statistic with a known sampling distribution, measuring closeness of implied covariances to sample covariances; can test null that $RMSEA < 0.05$ (“close” fit)

Model Fit Criteria

- **AIC / BIC:** Used to compare models with different counts of parameters, and no nesting relationship; not useful on their own; **smaller values indicate better fits**, with penalties applied for more parameters estimated
- **Likelihood ratio tests:** When ML estimation is used, standard LRTs can be used to compare the fits of nested models; good for simplifying model structures

Modification Indices

- If a model does not appear to fit well, most software can compute **modification indices**
- These provide a sense of how much the overall Chi-square statistic can be *reduced* by estimating additional parameters in the model
- These indices should **not** be used for exploratory analysis; they should only be used to make slight tweaks to a theoretically supported model

Alternative Estimation Methods

- **WLS/Asymptotic Distribution Free (ADF);** fitting function is written as follows:

$$F_{ADF} = (s - \sigma(\theta))'W^{-1}(s - \sigma(\theta))$$

where

s = vector of non-duplicated elements of S

$\sigma(\theta)$ = vector of implied variances and covariances

W = estimated variance-covariance matrix for elements of S

- W is a positive-definite “weight matrix”
- **Does not require multivariate normality;** W is based on “covariances of covariances”

Alternative Estimation Methods

- Categorical Variable Model (**CVM**); good for binary and ordinal variables; uses the general ADF function; uses *polychoric* correlations of underlying latent variables as inputs to ADF
- **Bootstrapping**: Another approach to dealing with non-normality; compute standard errors and correct Chi-square fit statistics based on many (> 500) bootstrap (with-replacement) samples from the observed data

Power and Sample Size

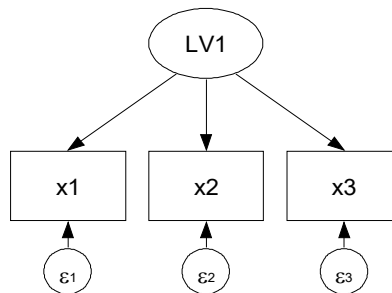
- In general, more cases are needed for:
 - Complex models (= many parameters)
 - Models with weak relationships in the population
 - Non-normal distributions
- At a minimum, one should use 100-150 cases
- Rule of thumb: 10-20 cases **per parameter estimated** (Kline, 2015, *Principles and Practice of SEM, Fourth Edition*)

Power and Sample Size

- More formal power calculations can be performed by simulation:
 - https://r-causal.github.io/ggdag/reference/simulate_data.html
 - <https://robindenz1.github.io/simDAG/index.html>
- Also, there are reference tables found in MacCallum et. al's (1996) paper "Power analysis and determination of sample size for covariance structure modeling"
- The first step in performing a power analysis for an SEM is drawing the conceptual model and then calculating the degrees of freedom (see **Slide 17**)

Example: Confirmatory Factor Analysis (CFA)

- Suppose that we have three observed MVN variables, and theory suggests that they indicate one latent construct (LV1):



Example: Confirmatory Factor Analysis (CFA)

- We **scale** the latent factor to have the same measurement as the first observed variable by fixing the first path (coefficient) to 1
- Equations:

$$x_1 = v_1 + LV_1 + \varepsilon_1$$

$$x_2 = v_2 + \lambda_{21}LV_1 + \varepsilon_2$$

$$x_3 = v_3 + \lambda_{31}LV_1 + \varepsilon_3$$

Example: Confirmatory Factor Analysis (CFA)

- **Model-Implied Variances and Covariances:**

$$\Sigma(\theta) = \begin{bmatrix} \text{Var}(LV_1) + \text{Var}(\varepsilon_1) & & \\ \lambda_{21}\text{Var}(LV_1) & \lambda_{21}^2\text{Var}(LV_1) + \text{Var}(\varepsilon_2) & \\ \lambda_{31}\text{Var}(LV_1) & \lambda_{21}\lambda_{31}\text{Var}(LV_1) & \lambda_{31}^2\text{Var}(LV_1) + \text{Var}(\varepsilon_3) \end{bmatrix}$$

- **Model-implied Means:**

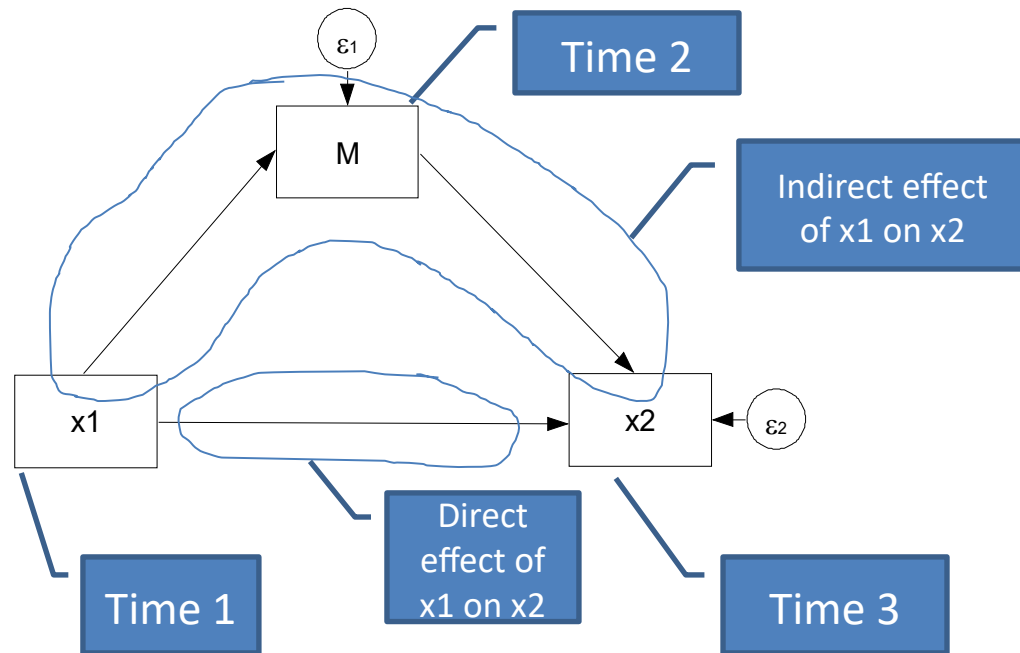
$$\mu(\theta) = \begin{bmatrix} v_1 + \text{Mean}(LV_1) \\ v_2 + \lambda_{21}\text{Mean}(LV_1) \\ v_3 + \lambda_{31}\text{Mean}(LV_1) \end{bmatrix}, \text{ with } \text{Mean}(LV_1) \text{ known, given scaling}$$

- These pieces are input to the fitting function (**Slide 15**) for ML estimation of the 9 unknown parameters (can you see the unknown parameters? $v_1, v_2, v_3, \text{var}(LV_1), \text{var}(\varepsilon_1), \text{var}(\varepsilon_2), \text{var}(\varepsilon_3), \lambda_{21}, \lambda_{31}$)
- 6 unique variance/covariance parameters + 3 means = 9 inputs; 9 parameters estimated $\rightarrow$ 0 DF (**saturated model**: we are exactly replicating the observed correlations, but our interest is in the estimated coefficients!)

Example: Path Model

- Suppose that we are interested in testing a **mediation model**
- An apparent **direct effect** of one observed variable (x_1) on another (x_2) may be due to the **indirect effect** of x_1 on x_2 through a third variable (a mediator, M)
- In a path model, **all variables are observed!**
- In mediation models, temporal ordering of variables in causal pathway is important!

Example: Path Model



Total Effect of x_1 on x_2 = direct effect + indirect effect

Example: Path Model

- **Equations:**

$$x_2 = v_2 + \beta_{21}x_1 + \beta_{2M}M + \varepsilon_2$$

$$M = v_M + \beta_{M1}x_1 + \varepsilon_1$$

- **Model-Implied Variances and Covariances:**

$$Var_{\theta} \begin{bmatrix} M \\ x_2 \\ x_1 \end{bmatrix} = \Sigma(\theta) = \begin{bmatrix} \Sigma_{endo}(\theta) & \Sigma_{endo,exog}(\theta) \\ \Sigma_{endo,exog}(\theta) & \Sigma_{exog}(\theta) \end{bmatrix} = \begin{bmatrix} [\Gamma\Phi\Gamma' + \Psi] & \Gamma\Phi \\ \Phi\Gamma' & \Phi \end{bmatrix}$$

where

Γ = vector of (5) coefficients in equations for endogenous variables (M and x_2)

Φ = variance-covariance matrix of exogenous variables (1 element, known)

Ψ = variance-covariance matrix of errors for endogenous variables (2 elements)

Example: Path Model

- **Model-Implied Means:**

$$\mu(\theta) = \begin{bmatrix} v_M + \beta_{M1}Mean(x_1) \\ v_2 + \beta_{21}Mean(x_1) + \beta_{2M}Mean(M) \\ Mean(x_1) \end{bmatrix}$$

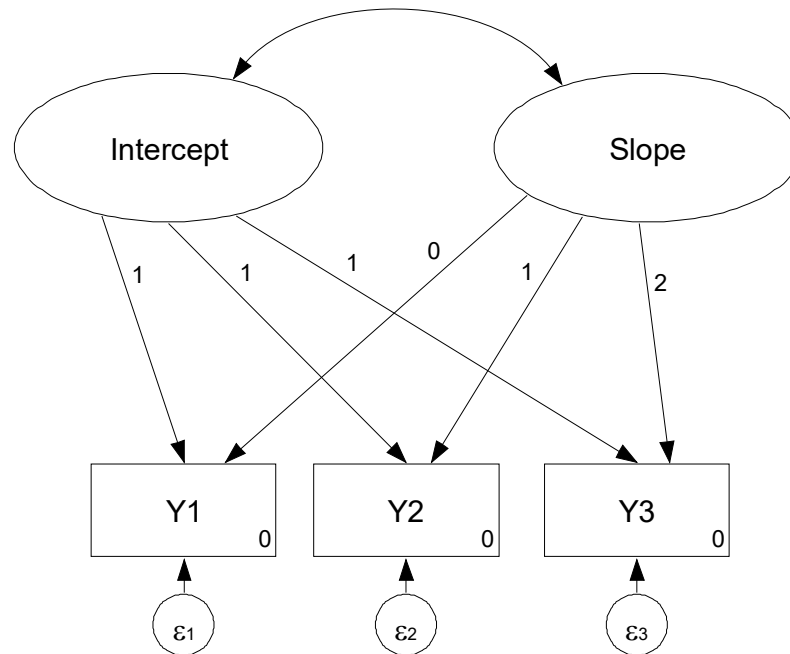
- **Inputs:** 3 variances, 3 covariances, 1 *unique* mean (the means of the endogenous variables, x2 and M, are exactly defined by known information and other parameters in this model)
- **Parameters to be estimated:** 7
- **Degrees of Freedom:** 0 (a **saturated model**, meaning that this model perfectly replicates the observed correlations; that's OK, because we can still generate our parameter estimates and their SEs!)

Example: Latent Growth Curve Model (LGCM)

- Multilevel models for longitudinal data can also be fitted using an SEM framework
- In this case, we treat random intercepts and slopes as latent variables, and estimate their means (**fixed effects!**) and (co)variances (**D!**)
- We have to initially fix paths from the latent intercepts and slopes to represent changes in units of time across the observed variables

Example: LGCM

- An example, with time points 0 (baseline), 1 unit later, and 2 units later (“intercepts” = 0):



- **Equation:** $Y1_i = 0 + \text{Intercept_i} + 0 \times \text{Slope_i} + e1_i$

Example: LGCM

- The fact that multilevel models can be fitted in the SEM framework opens up many interesting modeling possibilities
- One has to be careful to stick to theoretical considerations, however, as SEM models can become quite complicated!
- **Also important:** note that a **wide** data structure is required for fitting these SEMs

Visualizing SEM Fits

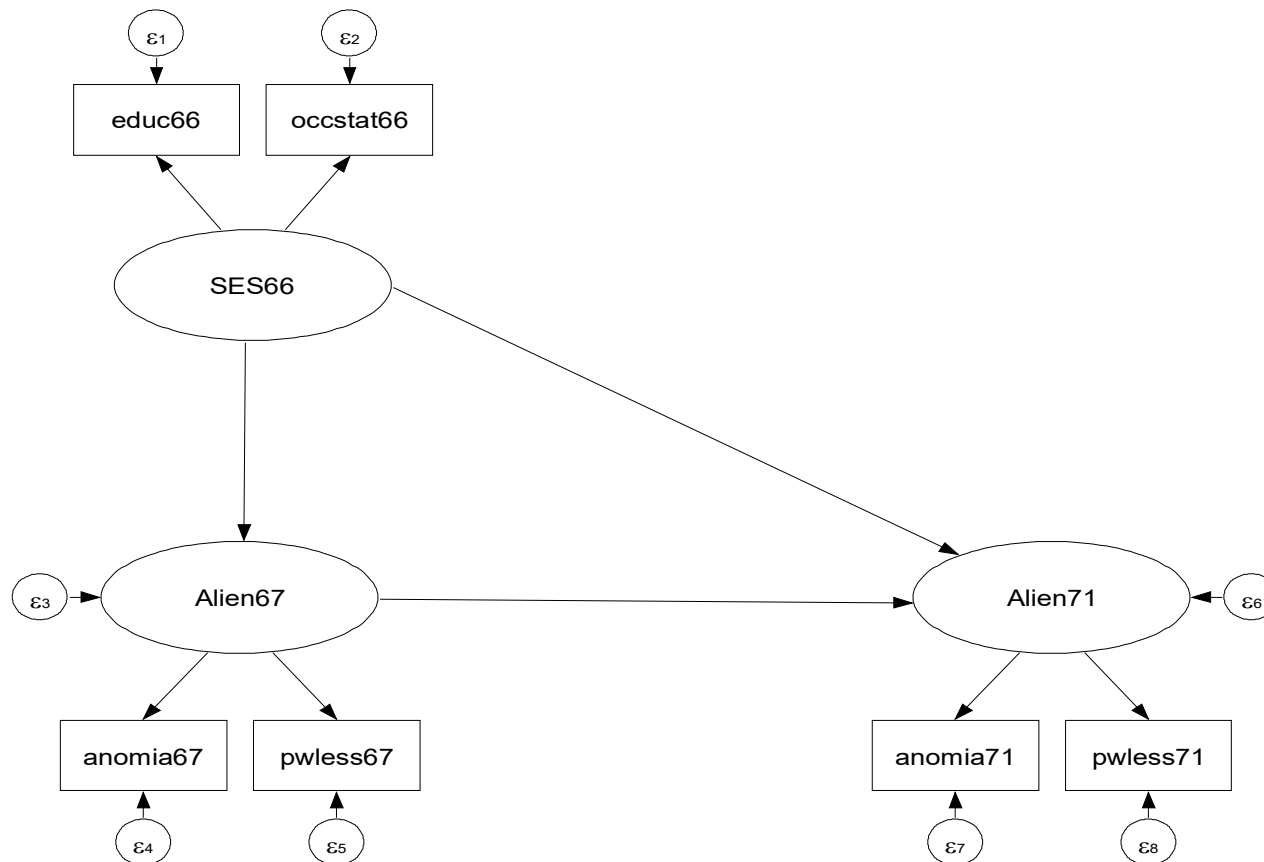
- See **Hallgren et al. (2018)** on Canvas
- Simple figures (e.g., path models / SEMs shown earlier) illustrating SEM models that were fitted don't always convey the nature of the relationships modeled!

SEM Software Options

- R: The `sem` and `lavaan` packages
- SAS: `proc calis`
- Stata: `sem` / `gsem` commands
 - Models can be drawn or estimated using syntax (or both!)
- `gllamm` add-on command in Stata: syntax only
- AMOS (SPSS Add-On Module)
 - Models are drawn and then estimated
- Mplus: Syntax only
- RLISREL
- EQS
- We will consider examples in R, Stata and Mplus (mainly FYI)

Example R Code

- Recall the SEM on **Slide 5** (data on Canvas):



Example R Code

```
install.packages("lavaan", dependencies=TRUE)
library(lavaan)
wheaton <- read.table("wheaton.dat", h=T) # if there is a header (DATA ON CANVAS)
model <- ' # CRITICAL: NOTE SINGLE QUOTES USED TO DEFINE MODEL OBJECT
# measurement model
    ses66 =~ educ66 + occstat66
    alien67 =~ anomia67 + pwless67
    alien71 =~ anomia71 + pwless71
# structural model
    # direct effect
        alien71 ~ c*ses66
    # mediator
        alien67 ~ a*ses66
        alien71 ~ b*alien67
    # indirect effect (a*b)
        ab := a*b
    # total effect
        total := c + (a*b)
    ,
fit <- sem(model, data=wheaton)
summary(fit)
```

SEE TODAY'S HANDOUT FOR DETAILED OUTPUT

Thinking Ahead

- In the next class, we will consider important extensions of the SEM framework:
 - Multiple Group Analysis (MGA)
 - Latent Class Analysis (LCA)
- LCA is often used for measurement error investigations in surveys; we'll see an example
- Later, we will also consider how to account for complex sampling features when fitting SEMs and making inferences